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Rankine Cycle Derivations

In this part, we consider a four control volume system representing a simple Rankine cycle. This will form the theoretical basis to

Chapter 1

Mass, Energy and Momentum Balance

In OpenFOAM, the PIMPLE algorithm is often used to solve for incompressible fluid flow [Devolder et al., 2015], this is one of the many standard solvers for solving flow in OpenFOAM [Penttinen, Yasari, and Nilsson, 2011]. Jasak has published this in his PhD thesis, available online, [Jasak, 1996] on page 143-149.

1.1 Raw forms of mass, momentum and energy balance

Let's consider mass balance and momentum balance:

$$\frac{\partial \dot{m}}{\partial t} = \sum_i m_i \tag{1.1}$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla P + \nabla \cdot \tau + \rho g \tag{1.2}$$

$\nabla \cdot \tau$ represents shear, ρg represents the gravity body force, $-\nabla P$ represents the pressure gradient.

$$V_j \frac{\partial \rho_i h_i}{\partial t} + \sum_j \dot{m}_j h_j = Q_i - W_i + \sum_j \tau_j \cdot \mathbf{u}_j \tag{1.3}$$

Where $+\sum_j \tau_j \cdot \mathbf{u}_j$ is basically the friction loss terms where energy is transformed from kinetic energy to sensible heat. Q and W represent the thermodynamic work

1.2 converting forms to mass flowrate

It is often more convenient to convert the momentum equations to be in terms of mass flowrates. This is for 1D case.

To do so, we transform this by consider this 3D equation.

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla \cdot (\rho\mathbf{u}\mathbf{u}) = -\nabla P + \nabla \cdot \tau + \rho g \tag{1.4}$$

We need to consider just the x component, for a simple 1D case.

Then multiply by the cross sectional area.

1.3 Approach to Solving nonlinear Equations

Jasak in his PhD thesis outlines the methodology for solving this set of equation [Jasak, 1996] on page 143-149. These were the foundations for what would eventually become OpenFOAM Jasak, Jemcov, Tukovic, et al., 2007 licensed under GPLv3.

The momentum equation is quadratic in u, which makes the equation solution nonlinear. Solving this nonlinear equation would either need a nonlinear solver such as Newton Raphson method, or solving a linearised form of these equations, and iterating to convergence. The latter was chosen [Jasak, 1996].

$$\nabla \bullet (\mathbf{u}\mathbf{u}) = \sum_f \mathbf{S} \cdot \mathbf{u}_f \mathbf{u}_f = \sum_f F\mathbf{u}_f = a_p\mathbf{u}_p + \sum_N a_N\mathbf{u}_n \tag{1.5}$$

Finite Volume Discretisaion, Face and Centroid Properties

Section 3.3.1 within Jasak's thesis explains these terms.

First, we need to understand that Gauss's theorem is used in Finite Volume analysis, which is the bedrock of OpenFOAM solvers. $\sum_f \mathbf{S} \cdot \mathbf{u}_f \mathbf{u}_f$ represent the surface vector $\mathbf{S}$ and this will have a dot product with the velocity at the faces of the control volume $\mathbf{u}_f$.

Note that the subscript f represents values at the faces.

$\sum_f \mathbf{S} \cdot \mathbf{u}_f \mathbf{u}_f$ can be written in shorthand $\sum_f F\mathbf{u}_f$.

Now, we may not have the values of the velocities or fluxes at the faces. So we have to estimate it based on the properties at the centroid of the cell denoted p, and those of the neighbours denoted n.

Thus, the surface values denoted f are estimated based on the values at the cell centroids p and n based on some kind of interpolation scheme. This can be linear interpolation, or some upwind differencing scheme. This is described in page 81 of Jasak's thesis [Jasak, 1996].

With linear interpolation at the face, the solution can become unbounded. With upwind differencing, the solution is bounded, but diffusive. Much of the discussion revolves around how to estimate properties at the faces. This is why a whole host of differencing schemes comes up. Eg. at page 110 of Jasak's thesis [Jasak, 1996].

Linearisation and Predictor values

Given a very nonlinear system of equations. It was desirable to consider solving a linearised version of the momentum equations. We want to solve for the velocity $\mathbf{u}^{t+\Delta t}$ at the next timestep. But we only know $\mathbf{u}^t$.

We know that the face fluxes should be based on values in the cell centroids for both the cell p and that of the neighbouring cell n, hence we use:

$$a_p \mathbf{u}_p + \sum_N a_N \mathbf{u}_n$$

However, we have $\mathbf{uu}$ terms. To solve this problem and linearise this, we can use $\mathbf{u}^t \mathbf{u}^{t+\Delta t}$. Basically, $\mathbf{u}^t$ is known. This is the velocity at the previous timestep, and is therefore a constant. This will then help form the respective a_p and a_n constants so that the equations are linear.

Let's go back to our equations:

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{uu}) = -\nabla P + \nabla \cdot \tau + \rho g \tag{1.6}$$

In the incompressible assumption,

$$\frac{\partial(\mathbf{u})}{\partial t} + \nabla \cdot (\mathbf{uu}) = -\nabla P/\rho + \nabla \cdot \tau/\rho + g \tag{1.7}$$

We can do time discretisation:

$$\frac{\mathbf{u}^{t+\Delta t} - \mathbf{u}^t}{\Delta t} + \nabla \cdot (\mathbf{uu}) = -\nabla P/\rho + \nabla \cdot \tau/\rho + g \tag{1.8}$$

Now, remember, based on divergence theorem we need fluxes or properties at the faces:

$$\frac{\mathbf{u}^{t+\Delta t} - \mathbf{u}^t}{\Delta t} + \sum_f \mathbf{S} \cdot \mathbf{u}_f \mathbf{u}_f = -\nabla P/\rho + \nabla \cdot \tau/\rho + g \tag{1.9}$$

And since we cannot find properties at the faces, we have to estimate them using properties at the centroids.

Just to be clear, let's label the velocities at their proper locations, and then linearise the equation using some semi implicit time discretisation:

$$\frac{\mathbf{u}_p^{t+\Delta t} - \mathbf{u}_p^t}{\Delta t} + \sum_f \mathbf{S} \cdot \mathbf{u}_f^t \mathbf{u}_f^{t+\Delta t} = -\nabla P/\rho + \nabla \cdot \tau/\rho + g \tag{1.10}$$

The fluxes at the faces can be further estimated using some interpolation scheme. The easiest to understand is a linear scheme. Herein, we use some weighting factor to estimate the flux at the face between the cell and the other cell.

$$\mathbf{u}_f^t = w\mathbf{u}_p^t + (1-w)\mathbf{u}_n^t$$

Substituting this back:

$$\frac{\mathbf{u}_p^{t+\Delta t} - \mathbf{u}_p^t}{\Delta t} + \sum_f \mathbf{S}\cdot(w\mathbf{u}_p^t + (1-w)\mathbf{u}_n^t)(w\mathbf{u}_p^{t+\Delta t} + (1-w)\mathbf{u}_n^{t+\Delta t}) = -\nabla P/\rho + \nabla\cdot\tau/\rho + g \quad (1.11)$$

To make life easier for us, we can choose not to even bother with explicitly solving for the velocities of the neigbours

$$\frac{\mathbf{u}_p^{t+\Delta t} - \mathbf{u}_p^t}{\Delta t} + \sum_f \mathbf{S}\cdot(w\mathbf{u}_p^t + (1-w)\mathbf{u}_n^t)(w\mathbf{u}_p^{t+\Delta t} + (1-w)\mathbf{u}_n^t) = -\nabla P/\rho + \nabla\cdot\tau/\rho + g \quad (1.12)$$

Messy as the equation is, we now have a linear equation in velocities for each centroid at the next timestep $u_p^{t+\Delta t}$ and $u_n^{t+\Delta t}$.

We can then simplify this equation as such:

$$a_p\mathbf{u}_p^{t+\Delta t} = \mathbf{H}(\mathbf{u}^t) - \nabla P/\rho + \nabla\cdot\tau/\rho + g \quad (1.13)$$

$$\mathbf{u}_p^{t+\Delta t} = \frac{\mathbf{H}(\mathbf{u}^t)}{a_p} - \frac{\nabla P/\rho}{a_p} + \frac{\nabla\cdot\tau/\rho}{a_p} + g/a_p \quad (1.14)$$

Of course, the estimates for $\mathbf{u}^{t+\Delta t}$ are not going to be accurate. They are initial guess or **predicted** values.

We need to do some further correction to these predicted values in order to arrive at the correct answer.

Corrector steps

So far, we've only bothered with the momentum equation and ignored mass balance (or continuity in this case). Thus, there is no guarantee that the mass balance is fulfilled.

The mass balance is:

$$\nabla\cdot\mathbf{u}_p = 0$$

We then apply Gauss's divergence theorem to obtain:

$$\nabla\cdot\mathbf{u}_p = \sum_f \mathbf{S}\cdot\mathbf{u}_f = 0$$

Now again, how do we deal with the face fluxes again? We need to interpolate.

Suppose again that we can use some weighting factor (linear interpolation scheme), that on a face between a point and its neighbouring cell:

$$u_f = wu_p + (1-w)u_n$$

$$\mathbf{u}_p^{t+\Delta t} = \frac{\mathbf{H}(\mathbf{u}^t)}{a_p} - \frac{\nabla P/\rho}{a_p} + \frac{\nabla \cdot \tau/\rho}{a_p} + g/a_p$$

since

$$\mathbf{u}_p^{t+\Delta t} = 0$$

Then:

$$\nabla \cdot \left(\frac{\mathbf{H}(\mathbf{u}^t)}{a_p} - \frac{\nabla P/\rho}{a_p} + \frac{\nabla \cdot \tau/\rho}{a_p} + g/a_p \right) = 0$$

Moving the pressure term to the right:

$$\nabla \cdot \left(\frac{\mathbf{H}(\mathbf{u}^t)}{a_p} + \frac{\nabla \cdot \tau/\rho}{a_p} + g/a_p \right) = \nabla \cdot \frac{P/\rho}{a_p}$$

This is equation 3.141 in page 146 of Jasak's thesis [Jasak, 1996]. Of course, τ and g were not included.

Now, if we apply divergence theorem on the H terms:

$$\nabla \cdot \left(\frac{\mathbf{H}(\mathbf{u}^t)}{a_p} \right) = \sum_f \mathbf{S}_f \cdot \left(\frac{\mathbf{H}(\mathbf{u}^t)}{a_p} \right)_f$$

We also need to discretise the pressure side

$$\nabla \cdot \frac{P/\rho}{a_p} = \sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p} \right)_f$$

With this discretisation, we obtain

$$\sum_f \mathbf{S}_f \cdot \left(\frac{\mathbf{H}(\mathbf{u}^t)}{a_p} \right)_f = \sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p} \right)_f$$

With the interpolation:

$$u_f = wu_p + (1-w)u_n$$

the matrices can be set up. This is equation 3.143 in Jasak's thesis [Jasak, 1996].

So basically, the continuity equation is combined with the momentum equation to give a combined corrector equation.

1.4 PISO Algorithm

Pressure Implicit Splitting of Operators (PISO).

This includes a few steps, outlined in Jasak's thesis [Jasak, 1996]:

Step 1

This is the momentum predictor step, we can predict a velocity field using the previous steps

$$\mathbf{u}_p^{predictor} = \frac{\mathbf{H}(\mathbf{u}^t)}{a_p} - \frac{\nabla P/\rho}{a_p} + \frac{\nabla \cdot \tau/\rho}{a_p} + g/a_p \tag{1.15}$$

Step 2

With this, we can formulate a new $\frac{\mathbf{H}(\mathbf{u}^t)}{a_p}$:

$$\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p} \tag{1.16}$$

With this, we can substitute into the mass balance (pressure equation):

$$\sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p}\right)_f = \sum_f \mathbf{S}_f \cdot \left(\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p}\right)_f$$

This allows for a new pressure field. Note that the gravity terms are missing because hydrostatic pressure is ignored.

$$\sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p}\right)_f$$

It is conceivable that this results in some small matrix where pressure fields are simultaneously solved.

This is the pressure solution within Jasak's thesis page 147 [Jasak, 1996].

Step 3

We now have a new pressure field based on mass balance:

$$\sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p}\right)_f$$

And a new predicted H field based on the initial predictor step:

$$\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p}$$

Now, the predictor H field is not guaranteed to allow continuity with the old velocity, but since it was used to generate a new pressure field using the continuity equation, it should be able to guarantee creation of a new velocity field which will satisfy continuity equation.

$$\mathbf{u}_p^{corrected} = \frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p} - \left(\frac{\nabla P/\rho}{a_p}\right)_{\text{corrected pressure field}} \tag{1.17}$$

repeated iterations

This forms the basis for iterating through both mass and momentum balances. This is one round of PISO prediction-correction.

We may iterate this several times until the residual is small between consecutive iterations.

Now, it must be noted that during PISO, we do NOT repeat step 1 to recalculate as mentioned in page 148 Jasak's thesis Jasak, 1996:

$$\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p}$$

The H term is kept constant through the whole correction sequence and changed in only the next momentum predictor [Jasak, 1996].

1.5 SIMPLE Algorithm

The SIMPLE algorithm is similar, except that it is solving for steady state solutions, and uses large time step.

The key difference is rather than skip Step 1 within loops, we repeat step 1 for momentum predictor. This is the key difference between PISO and SIMPLE. It is that we run the momentum predictor step every time in SIMPLE. In this regard, it is more important to re-calculate H in every iteration so that non-linearities are accounted for.

Hence, between iterations, there is no need to fully resolve pressure-velocity coupling. Which means between iterations, one does not need to actively ensure that they are solved simultaneously.

Step 1

Same as PISO, this is the momentum predictor step, we can predict a velocity field using the previous steps

$$\mathbf{u}_p^{predictor} = \frac{\mathbf{H}(\mathbf{u}^t)}{a_p} - \frac{\nabla P/\rho}{a_p} + \frac{\nabla \cdot \tau/\rho}{a_p} + g/a_p \tag{1.18}$$

However, before going to step 2, we

Step 2

Also same as PISO, we do the pressure equation step

With this, we can formulate a new $\frac{\mathbf{H}(\mathbf{u}^t)}{a_p}$:

$$\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p} \tag{1.19}$$

With this, we can substitute into the mass balance (pressure equation):

$$\sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p}\right)_f = \sum_f \mathbf{S}_f \cdot \left(\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p}\right)_f$$

This allows for a new pressure field. Note that the gravity terms are missing because hydrostatic pressure is ignored.

$$\sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p}\right)_f$$

It is conceivable that this results in some small matrix where pressure fields are simultaneously solved.

This is the pressure solution within Jasak's thesis page 147 [Jasak, 1996].

Step 3

We now have a new pressure field based on mass balance:

$$\sum_f \mathbf{S}_f \cdot \left(\frac{P/\rho}{a_p}\right)_f$$

And a new predicted H field based on the initial predictor step:

$$\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p}$$

Now, the predictor H field is not guaranteed to allow continuity with the old velocity, but since it was used to generate a new pressure field using the continuity equation, it should be able to guarantee creation of a new velocity field which will satisfy continuity equation.

$$\mathbf{u}_p^{corrected} = \frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p} - \left(\frac{\nabla P/\rho}{a_p}\right)_{\text{corrected pressure field}} \tag{1.20}$$

repeated iterations

This forms the basis for iterating through both mass and momentum balances. This is one round of SIMPLE prediction-correction.

We may iterate this several times until the residual is small between consecutive iterations.

Now, it must be noted that previously during PISO, we do NOT repeat step 1 to recalculate as mentioned in page 148 Jasak's thesis Jasak, 1996:

$$\frac{\mathbf{H}(\mathbf{u}^{predictor})}{a_p}$$

Now, in SIMPLE, we do change H during subsequent iterations in the momentum predictor step [Jasak, 1996].

Dr Aiden's YouTube videos were helpful in understanding this regard too.

Relaxation Factors

However, between each iteration, timesteps may be large. Hence, the solution may experience some form of instability. Therefore, there is the introduction of relaxation factors which updates the velocity slowly between successive momentum predictor steps.

Jasak mentions the use of velocity under-relaxation factors α_u. as well as pressure under-relaxation factors α_p.

He outlined the following steps on page 148-149 of his thesis [Jasak, 1996]. For pressure:

$$p^{new} = p^{old} + \alpha_p(p^{predictor} - p^{old})$$

For momentum, a similar thing applies. Though for momentum α_u is usually around 0.8 and for pressure α_p is around 0.2. Jasak writes this in page 149 of his thesis [Jasak, 1996].

1.6 PIMPLE Algorithm

For the PIMPLE algorithm, this is a merger between PISO and SIMPLE algorithms [Greenshields and Weller, 2022].

Basically, every SIMPLE iteration now has the option to perform a set number of PISO iterations as well. That is perform the corrector steps multiple times per SIMPLE iteration. The PISO loops are known as inner loops, whereas the outer loops are known as SIMPLE loops.

It is then up to the user to know how many loops are required.

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